D02 · S&OP; / IBP Governance

# 03 · Scenario planning & what-if analysis in S&OP;

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## 1 WHAT IT IS

Scenario planning in S&OP carries more than one future through the cycle — best/base/worst demand, supply disruptions, and financial sensitivities — with pre-defined triggers and contingency responses, so the organization decides how it will react before the deviation occurs rather than improvising when it does.

## 2 WHY IT MATTERS

Schoemaker 1995 **(Schoemaker, 1995 — peer-reviewed)** argued that scenario planning's value is not predicting the future but expanding the set of futures an organization is prepared to act on, reducing both overconfidence and blind spots. An S&OP that approves a single point plan has implicitly bet the business on one forecast being right; when demand swings or a supplier fails, there is no pre-agreed response, so the organization burns its first and most valuable days debating what to do instead of executing. The cost of that delay is concrete: **the response window for a disruption is widest the day it is detected and narrows fast**, and an unprepared organization spends it deciding rather than acting.

## 3 HOW IT IS DONE

1. **Define a small set of decision-relevant scenarios.** Not dozens — three to five that would actually change a decision: a demand upside, a demand downside, a key-supplier disruption, a cost shock. Scenarios that do not change any action are analysis theater.
2. **Quantify each scenario through the plan, in units and dollars.** Run each future through demand, supply, and the financial bridge so leadership sees the operational and P&L consequence of each, not just a demand swing in isolation.
3. **Pre-decide the response for each scenario.** For each material scenario, agree the action now: which capacity flexes, which inventory pre-builds, which orders defer. The decision made in calm is faster and better than the one made in crisis.
4. **Set explicit triggers.** Define the leading indicator and threshold that activates each contingency — a demand signal, a supplier lead-time slip, a commodity price level. A response without a trigger never fires until it is too late.
5. **Size the cost of optionality.** Pre-built inventory and reserved capacity cost money; quantify the premium so leadership buys flexibility deliberately, not by default. Optionality is an investment with a price, not a free hedge.
6. **Review which scenarios materialized.** Each cycle, check which triggers fired and whether the pre-agreed response worked. Scenario planning that never reviews its own hit rate degrades into a ritual that produces slides no one acts on.

> **WORKED EXAMPLE — ILLUSTRATIVE**
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> An industrial firm approved a single base-case S&OP plan each month — **baseline: no scenarios, average 9 days from a demand or supply shock to an executed response, repeated expedite and overtime to recover**. It introduced three standing scenarios with triggers and pre-agreed responses. Results: **shock-to-response time 9 days -> 2 days, expedite spend on disruptions down ~30%, and a demand-downside scenario that pre-authorized a build pause avoided an estimated $2.4M in excess** when that trigger fired in Q3.

**IN PRACTICE**

**Keep the scenario set small and standing**

Maintain three to five standing scenarios reviewed every cycle rather than reinventing them monthly. A stable set builds organizational muscle memory; a shifting one produces fresh slides and no readiness. Add a scenario only when a new, decision-changing risk emerges.

### Rapid re-planning capability

Scenario value depends on being able to run a future through the full plan quickly. Invest in a planning model that re-computes demand-supply-finance in minutes, not the days a manual rebuild takes — the speed of the what-if is the speed of the decision.

### Documented contingency playbooks

For each material scenario, hold a one-page playbook: trigger, owner, first actions, and the decisions already pre-authorized. When the trigger fires, the playbook turns a crisis meeting into the execution of a plan that already has sign-off.

### Trigger-hit review

Each cycle, review which triggers fired, which responses executed, and how well they worked. This closes the learning loop and keeps scenario planning from decaying into an unread appendix — the discipline of reviewing hit rate is what keeps the scenarios honest.

#### WHAT YOU WOULD USE

**SCENARIO / WHAT-IF PLANNING**

**Kinaxis / o9 — enterprise tier-1** — Concurrent planning engines built for fast scenario simulation across demand, supply, and finance in a single model. **(Kinaxis, o9 — vendor)** The choice when what-if speed across the full network is the differentiator — re-planning in minutes is the core capability.

**SAP IBP / Blue Yonder** — Solid scenario and simulation capabilities integrated with the planning baseline. **(SAP, Blue Yonder — vendor)** A strong fit where scenarios must reconcile with the committed S&OP plan rather than live in a separate model.

**Connected-planning / simulation specialists** — Flexible modeling platforms (e.g. Anaplan) and dedicated simulation tools strong at financial what-if. **(specialized vendors — vendor)** The vertical option when the scenario work is heavily financial and decoupled from the supply engine.

**For operations that don't need a platform** — A small set of standing scenarios modeled in a spreadsheet with documented triggers and one-page playbooks. The discipline of pre-deciding responses delivers most of the value; fast tooling helps, but a slow model with agreed playbooks still beats a fast model with none.

## 4 SO WHAT

**Operations that carry scenarios and triggers execute a pre-agreed response while the window is still open; operations that approve one point plan spend the disruption deciding what to do.** Scenario planning does not predict the future — it pre-pays the decision cost of the futures that matter. The single-point plan feels efficient right up to the moment the forecast is wrong, which it always eventually is.

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